

Interdisciplinary Project-Based Learning (IPBL) 2026

Project Name: Smart Green Tunnel Ecosystem: Kinetic Harvesting & Automation Subsystem

OUR TEAM



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PROBLEM STATEMENT

What is the Problem?

Subterranean transit tunnels are severe municipal power sinks because safety regulations mandate that lighting, air quality monitoring, and massive forced ventilation fans operate continuously 24/7.

Concurrently, megawatts of kinetic energy generated by the massive physical weight of heavy traveling vehicles are completely lost as mechanical friction, vibration, and heat dissipation into passive roadway surfaces.

RESEARCH

What is the Problem?

Running tunnels on standard grids results in massive municipal utility bills, high maintenance costs, and a heavy carbon footprint.

When and where does the problem occur?

In remote regions, power grid failures turn tunnels into immediate safety hazards due to sudden total darkness and toxic exhaust accumulation.

What are users complaining about the existing solutions?

Why is it a problem?

Running tunnels on standard grids results in massive municipal utility bills, high maintenance costs, and a heavy carbon footprint.

Who is facing the problem?

National Highway Authorities, municipal corporations burdened by utility bills, and daily commuters who face safety risks from poor lighting and lack of emergency backup systems.

What are the existing solutions?

- Drawing 100% of operational electricity from distant, coal-burning power grids.
- Standard structural solar arrays placed outside tunnel mouths

What are users complaining about the existing solutions?

Standard hardware harvesters are rarely paired with automated edge logic, meaning they don't use the traffic data to conserve power in real time.

EMPATHY MAP / USER JOURNEY MAP



- Continuous power consumption regardless of actual traffic demand.
- Increased operational costs and carbon emissions.
- Limited use of real-time occupancy-based control systems

OUR ITERATED IDEA

The Concept: A modular, flush-embedded "**Horizontal Interval Compression Strip**" placed perpendicular to the flow of traffic every 3 to 5 meters.

How it solves the stakeholder drawbacks:

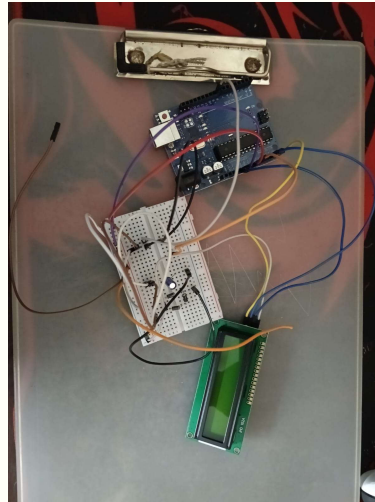
100% Capture Rate: The horizontal orientation ensures that every single vehicle tire strikes the system, completely eliminating the lane-drift errors of older longitudinal discs.

Dual-Purpose Transducer: The strip acts simultaneously as an energy scavenger and an algorithmic data-logging point.

PROTOTYPE



Product Brochure



- **Software:** Arduino IDE (Embedded C++ Platform)
- **Microcontroller:** ATmega328P Core (Arduino Uno R3)
- **Hardware Components:** 35mm Piezoelectric Ceramic Discs, Diodes, 10uF Electrolytic Smoothing Capacitor, 16x1 LCD Module, and Laser-Cut Acrylic Sheets.

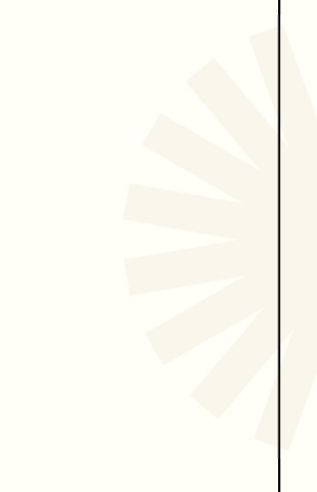
Evaluating the Prototype's Performance:

Testing & Validation

- Light Model (0.5 kg) @ 30 km/h → Generates 1.85V Peak → 3.42 mW Output
- Medium Model (1.2 kg) @ 45 km/h → Generates 3.10V Peak → 9.61 mW Output
- Heavy Model (2.5 kg) @ 60 km/h → Generates 4.85V Peak → 23.52 mW Output

BUSINESS MODEL CANVAS

Marketing Expenditure Category	Estimated Cost (INR)
High-Fidelity Industrial Boardroom Model	₹55,000
Cinematic 3D Explainer Video & Pitch Assets	₹50,000
B2B Infrastructure Expo Exhibition & Logistics	₹1,65,000
Localized On-Field Pilot Installation (Real-World Proof)	₹90,000
Total Estimated B2G Marketing Budget	₹3,60,000



**THANK YOU
SO MUCH**